

**Fair representation  
is integral to voting  
in a Democracy.**

**FAIR  
REPRESENTATION**

**Gerrymandering  
distorts the  
effect of an  
individual's vote.**

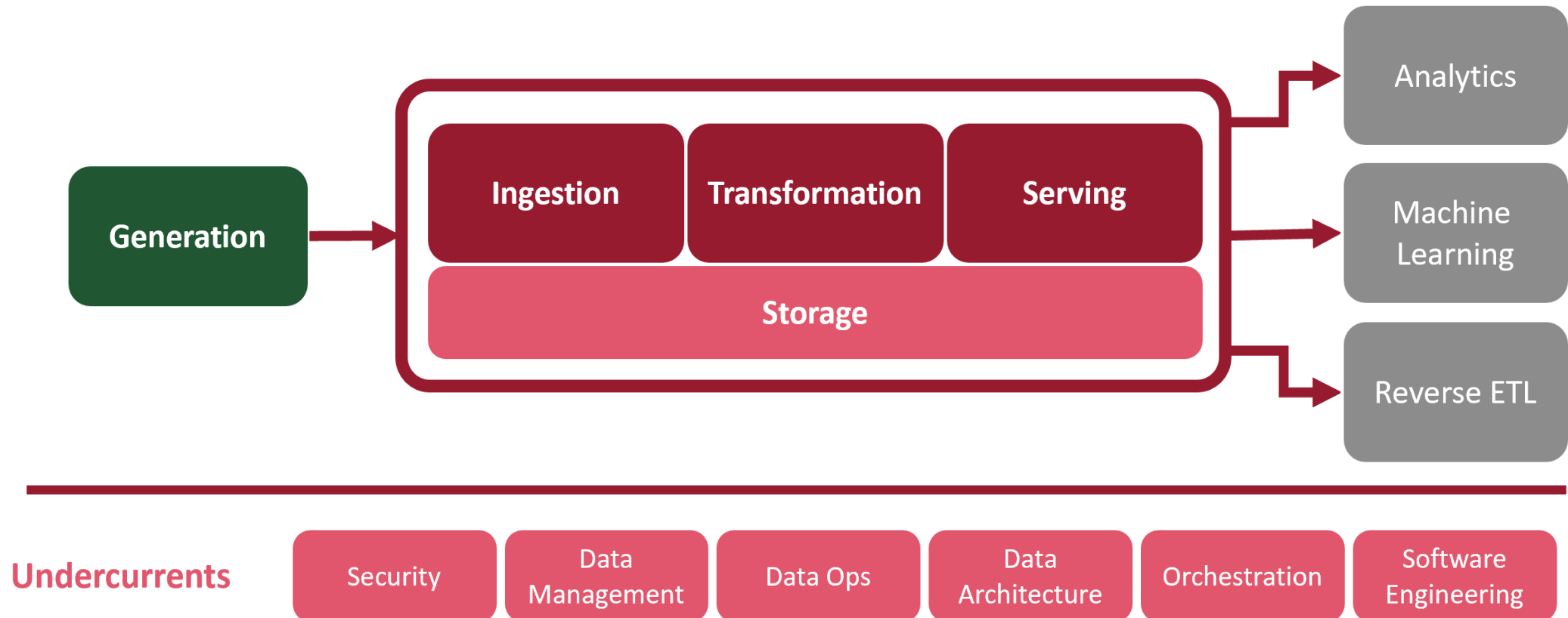
**GERRYMANDERING**



## **Identifying Partisan Gerrymandering Risk in U.S. Congressional Districts**

# Following The DE Lifecycle

The Data Engineering lifecycle will allow us to ingest data from various sources, calculate and refine features, and provide a service layer for end use & analytics.



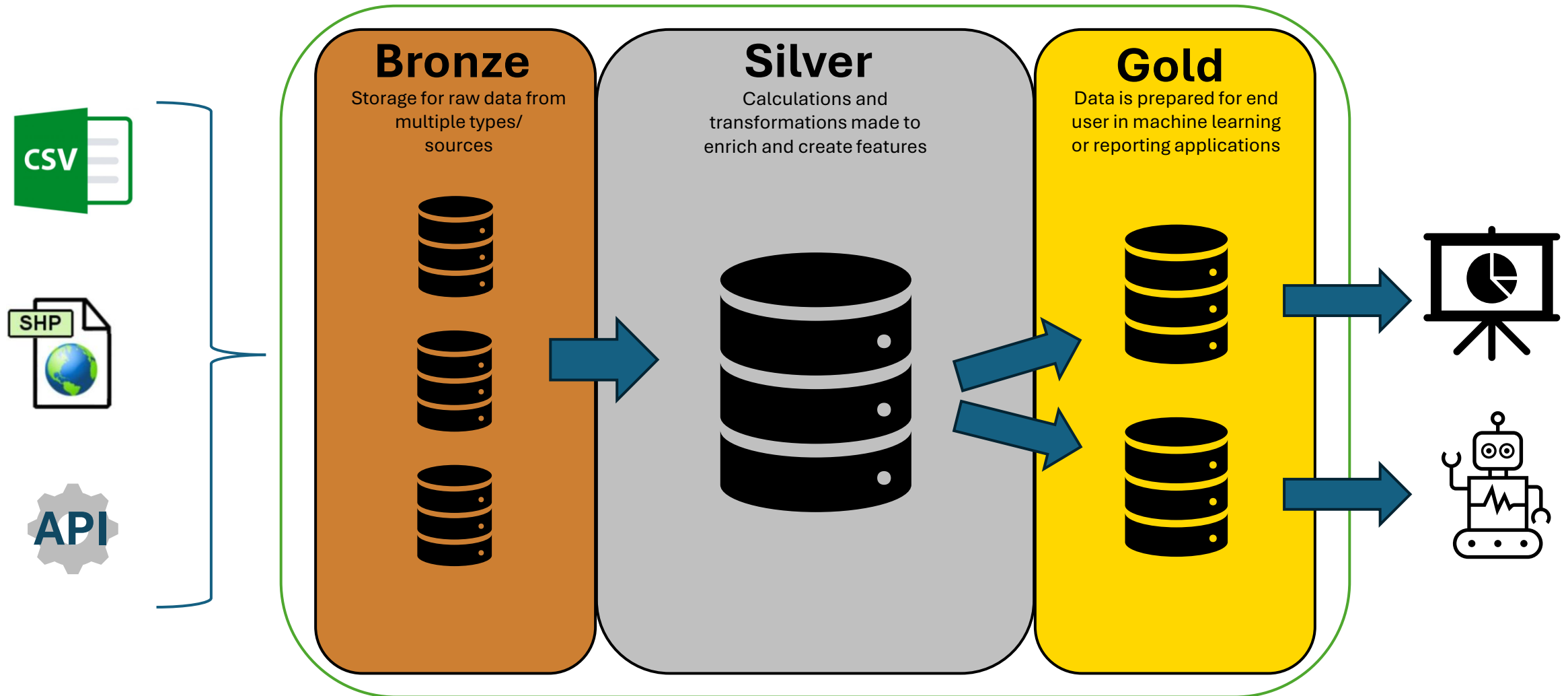
# Data Sources

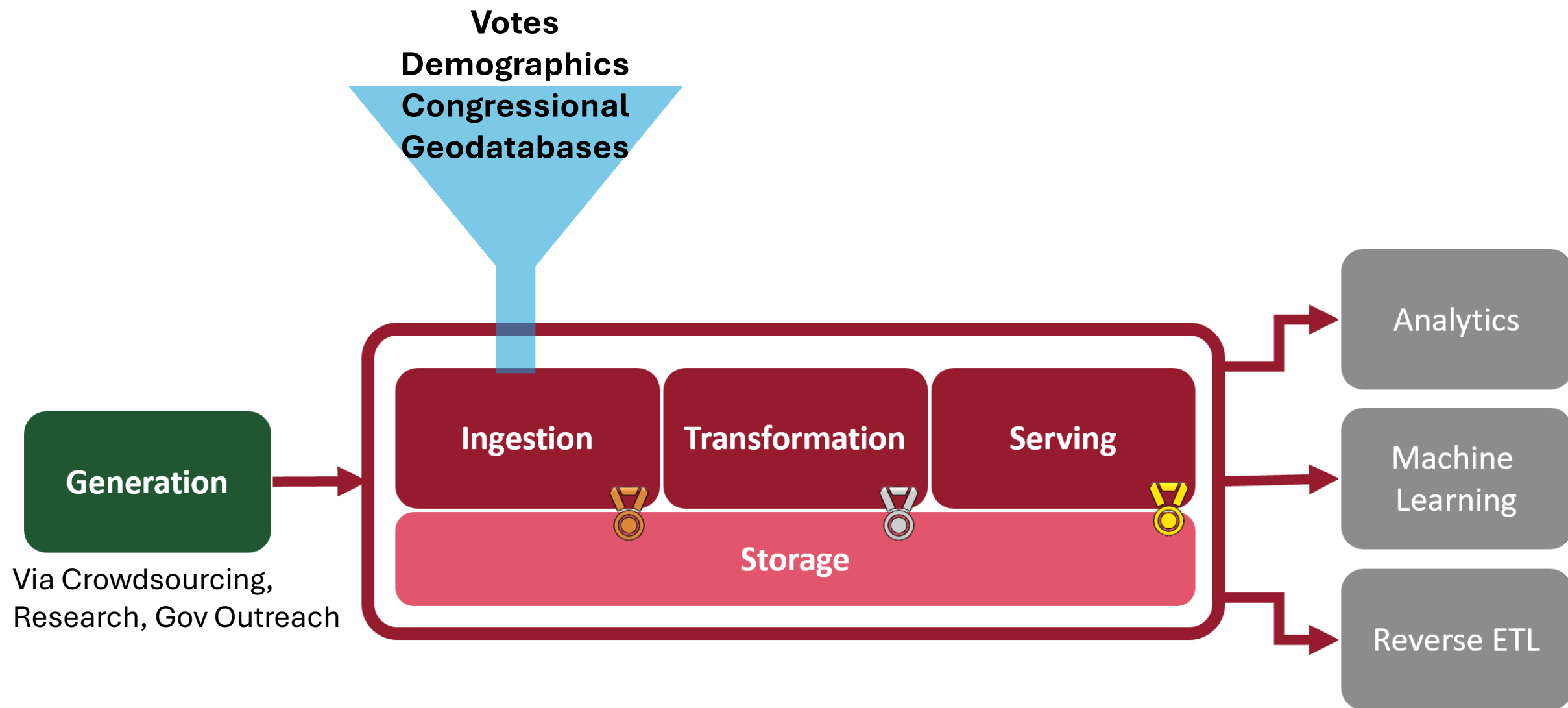
These sources focus on the key elements of our analysis: citizens and their voting data. Geospatial data serves as a plane underlying and connecting both.

| Data         | Source                                                                                                                                                                                                        | Summary                             | Documentation Process              | Accessibility       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------|---------------------|
| Voter Data   | <b>Open Elections</b><br><a href="https://openelections.net/">https://openelections.net/</a>                                                                                                                  | Precinct-level voter data           | Crowdsourced                       | GitHub (CSVs)       |
| Voter Data   | <b>Univ. of Florida Election Lab</b><br><a href="https://election.lab.ufl.edu/precinct-data/">https://election.lab.ufl.edu/precinct-data/</a>                                                                 | Voter turnout, counts, early voting | Faculty/student generated          | Website (Shapefile) |
| Demographics | <b>Census Bureau – American Community Survey</b><br><a href="https://www.census.gov/data/developers/data-sets/acs-5year.html">https://www.census.gov/data/developers/data-sets/acs-5year.html</a>             | Population, Race, Sex, Income, etc. | Federal government (via US Census) | API                 |
| Geographic   | <b>Census Bureau – REST Map Services</b><br><a href="https://tigerweb.geo.census.gov/tigerwebmain/TIGERweb_restmapservice.html">https://tigerweb.geo.census.gov/tigerwebmain/TIGERweb_restmapservice.html</a> | Land use, traits                    | Federal government                 | API                 |

# Medallion Architecture

Storage underlies the data sources, their transformations, and their final placement in a service layer.





# Transformation



Filtering and standardizing Open Elections for Congressional data

Calculating voting power, partisan symmetry

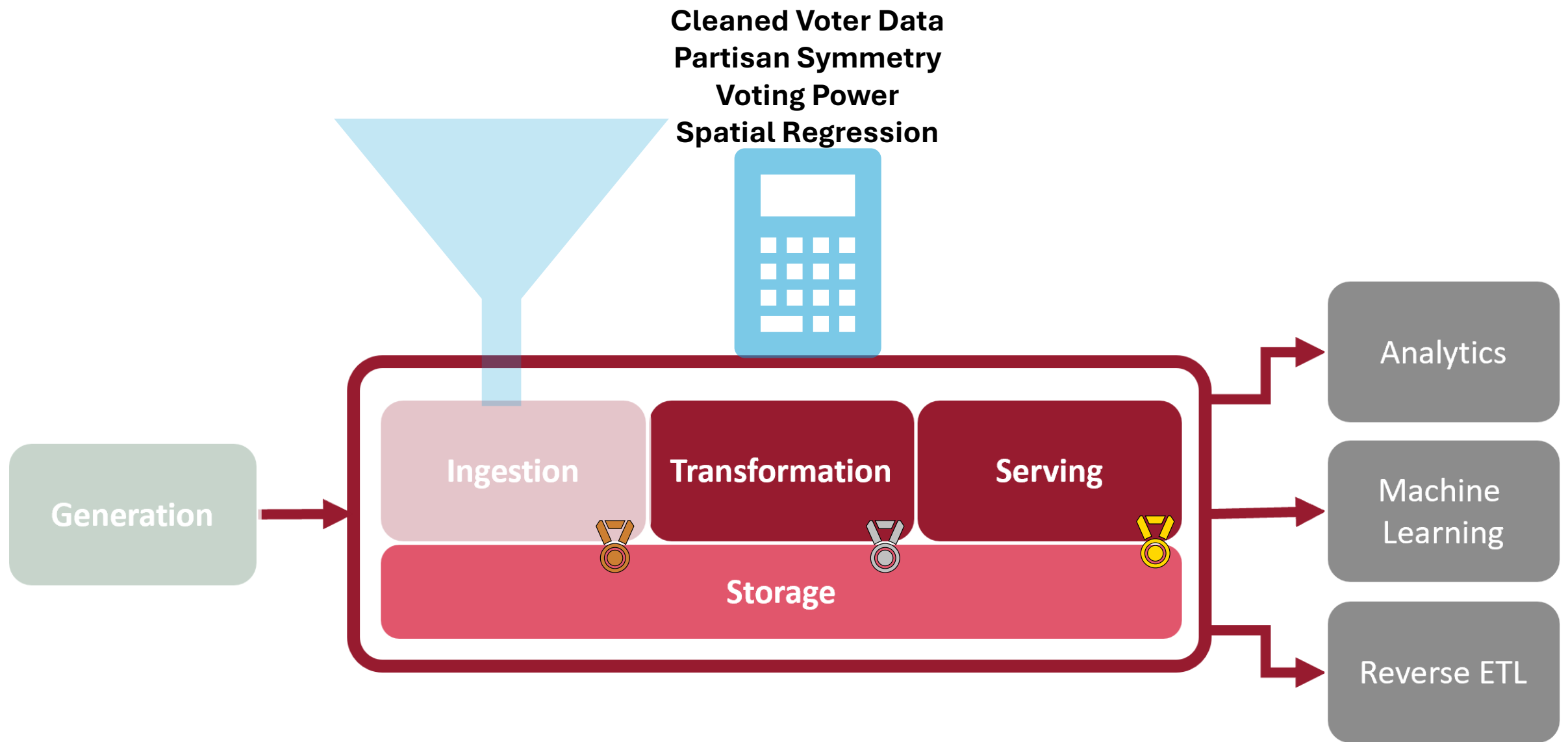
- Seats-Votes Curve, Efficiency Gap, Mean-Median Score

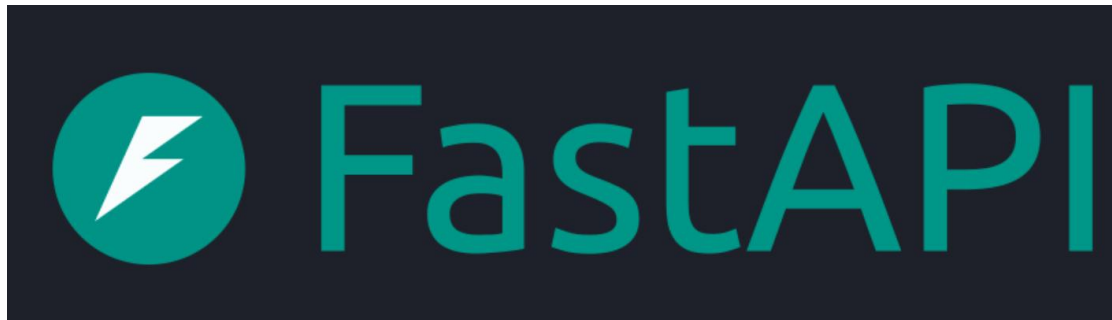
Demographics

- Distribution of populations, populations from precinct to Congressional District

Geospatial calculations

- Spatial Regression – geography as a feature
- Reock Score to measure compactness
- Boundaries (County, City, Precinct) present within a district, crossing districts
- Prior line control (partisan)





# Orchestration, Storage, Service

- Cloud storage for source data (Data Lake)
- Transform within cloud (Data Lakehouse)
- Serve transformed and raw data via API to users for analysis



# Thoughts on Teamwork...

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The team's biggest challenges will likely be the initial assembly of the components in our medallion architecture, and setting up orchestration



Completing this work can be done in agile sprints where each team member leans into their own strengths and provides deliverables regularly.

